

1.1 Physical Quantities & Units

Question Paper

Course	CIEA Level Physics
Section	1. Physical Quantities & Units
Topic	1.1 Physical Quantities & Units
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The average kinetic energy E of a gas molecule is given by the equation

$$E = \frac{3}{2} kT$$

where T is the absolute (kelvin) temperature.

What are the SI base units of k ?

- A. $\text{kg}^{-1} \text{m}^{-1} \text{s}^2 \text{K}$
- B. $\text{kg}^{-1} \text{m}^{-2} \text{s}^2 \text{K}$
- C. $\text{kg} \text{m} \text{s}^{-2} \text{K}^{-1}$
- D. $\text{kg} \text{m}^2 \text{s}^{-2} \text{K}^{-1}$

[1 mark]

Question 2

The speed of an aeroplane in still air is 200 km h^{-1} . The wind blows from the west at a speed of 85.0 km h^{-1} .

In which direction must the pilot steer the aeroplane in order to fly due north?

- A. 23.0° east of north
- B. 23.0° west of north
- C. 25.2° east of north
- D. 25.2° west of north

[1 mark]

Question 3

The units of all physical quantities can be expressed in terms of SI base units.

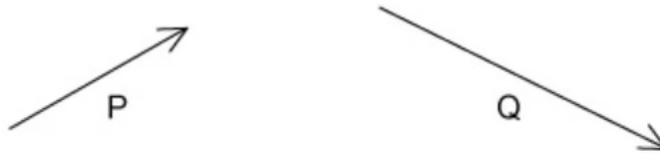
Which pair contains quantities with the same base units?

- A. Force and momentum
- B. Pressure and Young modulus
- C. Power and kinetic energy
- D. Mass and weight

[1 mark]

Question 4

Vectors **P** and **Q** are drawn to scale.



Which diagram represents the vector $(\mathbf{P} - \mathbf{Q})$?

A.

A



B.

B



C.

C



D.



[1 mark]

Question 5

The table contains some quantities, together with their symbols and units.

Quantity	Symbol	Unit
Gravitational field strength	g	N kg^{-1}
Density of liquid	ρ	kg m^{-3}
Vertical height	h	m
Volume of part of liquid	V	m^3

Which expression has the units of energy?

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Which expression has the units of energy?

A. $\rho g^2 h$

B. $\frac{\rho h V}{g}$

C. $\frac{\rho g}{h V}$

D. $gphV$

[1 mark]

Question 6

What is the ratio $\frac{10^{-3} \text{ THz}}{10^3 \text{ kHz}}$?

A. 10^{-9}

B. 10^{-6}

C. 10^0

D. 10^3

[1 mark]

Question 7

Which of the following correctly expresses the volt in terms of SI base units?

A. $A\Omega$

B. WA^{-1}

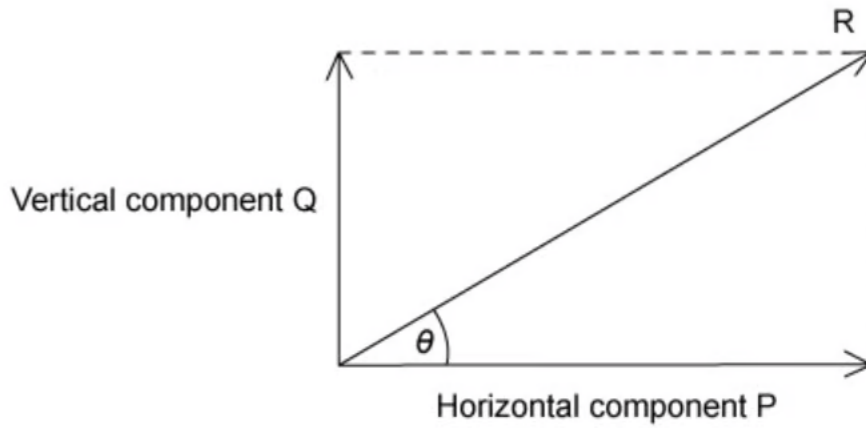
C. $kg \ m^2 \ s^{-1} \ A^{-1}$

D. $kg \ m^2 \ s^{-3} \ A^{-1}$

[1 mark]

Question 8

A vector has magnitude **R** and perpendicular components **P** and **Q**, as shown in the diagram.



Which row correctly describes the perpendicular components?

	Vertical component	Horizontal component
A	Q	$\sin \theta$
B	$R \cos \theta$	P
C	$R \cos \theta$	$R \sin \theta$
D	$R \sin \theta$	$R \cos \theta$

[1 mark]

Question 9

For which quantity is the magnitude a reasonable estimate?

- A. Frequency of a radio wave 500p Hz
- B. Mass of an atom 500μg
- C. The Young modulus of a metal 500 kPa
- D. Wavelength of green light 500 nm

[1 mark]

Question 10

The equation relating pressure and density is $P = \rho gh$

How can both sides of this equation be written in terms of base units?

- A. $[N\ m^{-1}] = [kg\ m^{-3}][m\ s^{-1}][m]$
- B. $[N\ m^{-2}] = [kg\ m^{-2}][m\ s^{-2}][m]$
- C. $[kg\ m^{-1}\ s^{-2}] = [kg\ m^{-3}][m\ s^{-2}][m]$
- D. $[kg\ m^{-1}\ s^{-1}] = [kg\ m^{-1}][m\ s^{-2}][m]$

[1 mark]

Model Answers: Medium

1

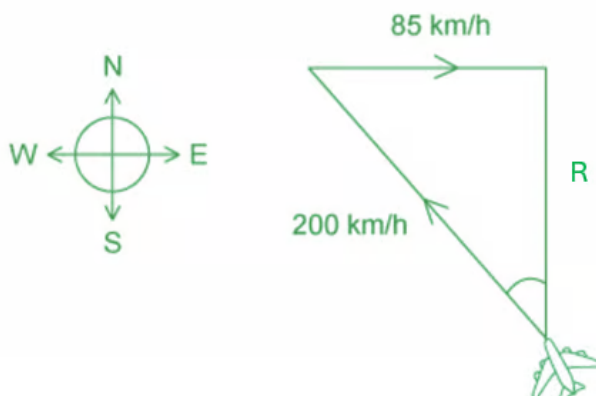
The correct answer is **D** because:

- We can ignore any numbers in the formula since they do not have any units.
- $k \propto \frac{E}{T}$ so, the unit will be along the lines of $J K^{-1}$
- We can reduce J to its SI unit by considering the equation for kinetic energy $E_k \propto mv^2$
 - $J = kg(ms^{-1})^2 = kgm^2s^{-2}$
- The units of k are
 - $JK^{-1} = kg m^2 s^{-2} K^{-1}$

2

The correct answer is **D** because:

- When we construct a vector triangle:
 - We know the wind acts from west to east
 - The resultant motion is due north, R
 - These two vectors connected tip to tail show
 - The direction required for the plane to move will be west of north



- Use trigonometry to calculate the angle from the north:

$$\circ \sin^{-1}\left(\frac{85}{200}\right) = 25.15^\circ$$

Try to picture a plane moving and being blown by the wind. Which way does it need to move to end up going north?

3

The correct answer is **B** because:

- Pressure is equal to

$$\circ P = \frac{F}{A} = Nm^{-2}$$

- Young Modulus is equal to

$$\circ Y = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$$

$$\circ \text{Stress: } \sigma = \frac{F}{A}$$

$$\circ \text{Strain: } \epsilon = \frac{x}{l}$$

$$\circ Y = \frac{\sigma}{\epsilon} = \frac{F}{A} \times \frac{l}{x} = \frac{N}{m^2} \times \frac{m}{m} = Nm^{-2}$$

Therefore, pressure and Young Modulus have the same base units.

A is incorrect as the base unit for force is $kg\ m\ s^{-2}$ and the base unit for momentum is $kg\ m\ s^{-1}$

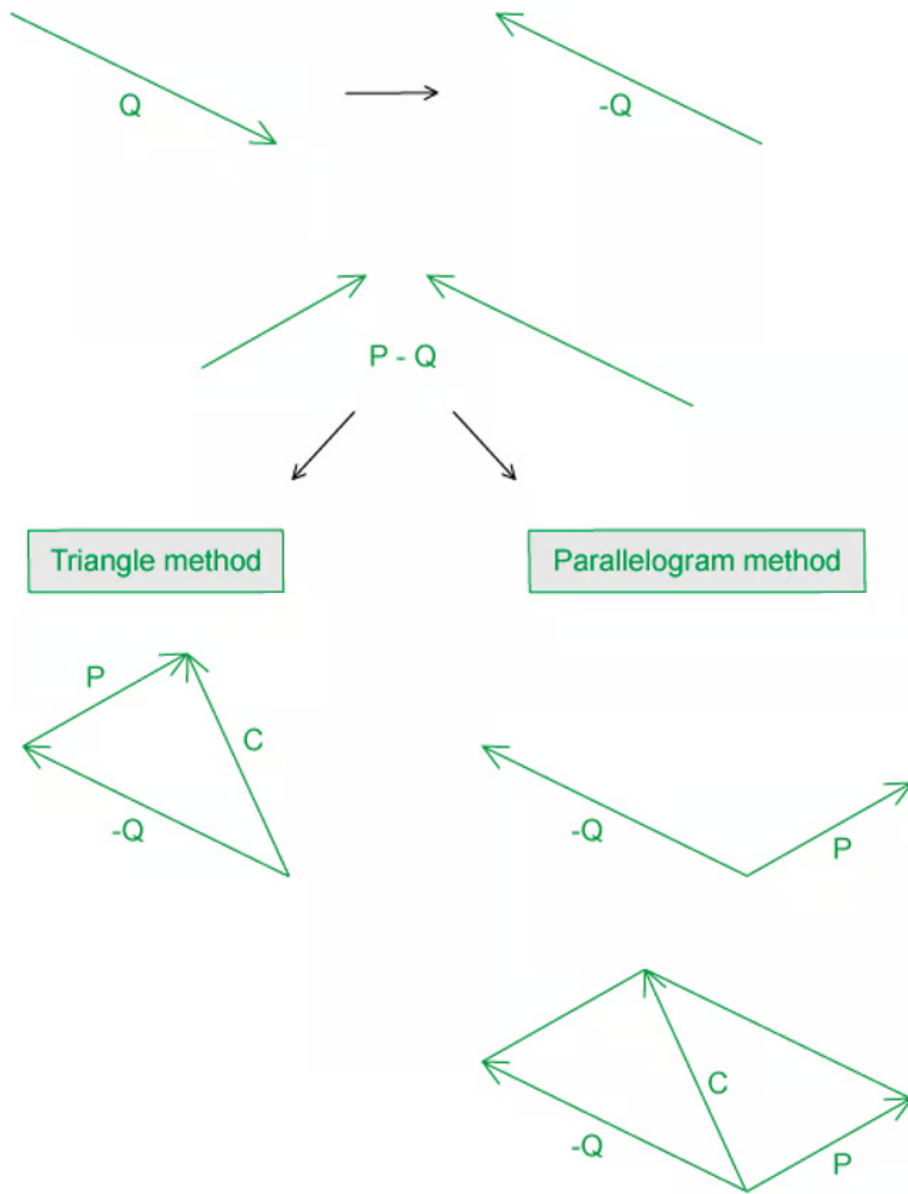
C is incorrect as the unit for power is $J\ s^{-1}$ and the unit for energy is J

D is incorrect as The base unit for mass is kg and the unit for weight is N or $kg\ m\ s^{-2}$

4

The correct answer is **C** because:

- First, it's important to notice that the vector **Q** is negative so its direction must be reversed
- Then using either the triangle or parallelogram method it's possible to determine the direction of the resulting vector, as shown in the diagram below



5

The correct answer is **D** because:

- The base unit of energy is $J = kg(ms^{-1})^2 = kg\ m^2\ s^{-2}$
 - $gphV = [ms^{-2}][kg\ m^{-3}][m][m^3]$

$$= N\ m$$

$$= (kg\ m\ s^{-2})m$$

$$= kg\ m^2\ s^{-2}$$
- Therefore, $gphV$ has the same units as energy

A is incorrect as the units of $\rho g^2 h = [kg\ m^{-3}][m\ s^{-2}]^2[m] = kg\ s^{-4}$

B is incorrect as the units of $\frac{\rho h V}{g} = [kg\ m^{-3}][m][m^3][m\ s^{-2}]^{-1} = kg\ s^2$

C is incorrect as the units of $\frac{\rho g}{h V} = [kg\ m^{-3}][m\ s^{-2}][m]^{-1}[m^3]^{-1} = kg\ m^{-6}\ s^{-2}$

6

The correct answer is **D** because:

- T = tera = 10^{12}
 - $10^{-3} \times 10^{12} = 10^{(12-3)} = 10^9$
- k = kilo = 10^3
 - $10^3 \times 10^3 = 10^{(3+3)} = 10^6$
- $\frac{10^9}{10^6} = 10^{(9-6)} = 10^3$

7

The correct answer is **D** because:

- Potential difference is equal to $\frac{\text{work done}}{\text{charge}}$
- So, the unit of the volt is $J\ C^{-1}$
- The base unit of energy $E = \frac{1}{2}mv^2$ is
 - $J = kg(ms^{-1})^2 = kgm^2s^{-2}$
- The base unit of charge $Q = It$ is
 - $C = As$
- So, the base unit of the volt is
 - $J\ C^{-1} = kg\ m^2\ s^{-2}\ A^{-1}\ s^{-1}$
- $V = kg\ m^2\ s^{-3}\ A^{-1}$

8

The correct answer is **D** because:

- Vertical component
 - $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{Q}{R}$
 - $Q = R \sin \theta$
- Horizontal component
 - $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{P}{R}$
 - $P = R \cos \theta$

9

The correct answer is **D** because:

- The range of visible light is about 400 – 700nm, so 500nm is a good estimate for the wavelength of green light
- You should have a rough idea of the ranges of each section of the electromagnetic spectrum

A is incorrect as with a frequency of 500pHz (500×10^{-12}), this would equate to a wavelength of $\frac{3 \times 10^8}{500 \times 10^{-12}} = 6 \times 10^{17} \text{ m}$.

Typical radio wavelengths range from 1 mm – 100 km so the estimated value is far too high

B is incorrect as in kg, 500μg works out as $500 \times 10^{-6} \times 10^{-3} = 5 \times 10^{-7} \text{ kg}$. The mass of a proton is $1.67 \times 10^{-27} \text{ kg}$ so the estimated value is far too high

C is incorrect as typical values for the Young Modulus of a metal are usually in the 50 - 400 GPa range so the estimated value is far too low

10

The correct answer is **C** because:

- Pressure is equal to
 - $P = \frac{F}{A} = \text{N m}^{-2} = (\text{kg ms}^{-2})\text{m}^{-2} = \text{kg m}^{-1} \text{ s}^{-2}$
- On the RHS we can combine units to give

- $[kg\ m^{-3}][m][m\ s^{-2}] = kg\ m^{-1}\ s^{-2}$

- The LHS matches the RHS so the correct answer is C